



What is autonomy?

The idea of autonomy is fundamental to the concept of agents but also a term that is very difficult to define, it can mean very different things to different people and so we will define a meaning that works well enough for our purposes.

Firstly, autonomy is a spectrum and at one extreme we have us, human beings, who are about as autonomous as you can get. We have control over our beliefs, actions and goals, or as much control as is possible. For example, I cannot simply choose to be able to fly and it happens, similarly I could not simply decide that $4+4$ equals 2. There are a lot of things that limit our possible choices whether that is genetics, which is preventing me from flying. Society also places restrictions on our choices, for example, we cannot simply take whatever we want from anyone unless we wish to suffer the consequences of the law. Years of human evolution and society norms mean that our behaviour is restricted but a human in the free world is about as autonomous as it gets because we set our own beliefs, goals and actions.

At the other end of the autonomy spectrum we might think of a software service, such as an API provided by a website. These services are not autonomous in any substantial way and they really just do what they are told. If you look at what you are using right now, a web browser, this cannot be considered in anyway autonomous as it just does what we tell it too.

However, in between these two extremes there is a range of possibilities. For example if we have a system that is operated via remote control, possibly in a dangerous environment that would be harmful for humans. The operator might observe how things are going and intervene where necessary but they agent goes about its business trying to reach its goals until that happens.

The points on the spectrum that we will be interested in with our agents is the point where we can delegate tasks to the system and then let the agent decide for itself the best way of handling things. The agent doesn't have complete autonomy like we might, it cannot decide which goals it wants to accomplish but it can decide how to accomplish them.